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MAULANA ABUL KALAM AZAD UNIVERSITY OF TECHNOLOGY, WEST BENGAL

Paper Code : M(CS)-401

NUMERICAL METHODS

Time Allotted : 3 Hours *Full Marks* : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

GROUP – A

(Multiple Choice Type Questions)

1. Choose the correct alternatives for any ten of the following : $10 \times 1 = 10$

   i) The number of significant digits in 1.00340 is

   a) 3 b) 4

   c) 5 d) 6.

   ii) The nth order forward difference of nth degree polynomial is

   a) n! b) (n+1)!

   c) 0 d) none of these.

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iii) Error in 4th order Runge-Kutta method is of the order of

a) h^2 b) h^3

c) h^4 d) none of these.

iv) The degree of precision of Simpson's 1/3rd rule is

a) 1 b) 2

c) 3 d) none of these.

v) In Trapezoidal rule, the portion of curve is replaced by

a) straight line b) parabolic path

c) circular path d) none of these.

vi) One root of the equation $x^2 + 2x - 2 = 0$ lies between

a) 1 and 2 b) 0 and 0.5

c) 0.5 and 1 d) none of these.

vii) The accuracy attainable with Newton-Raphson method does not depend upon the value of the derivative $f(x)$. The above statement is

a) True b) False

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viii) Gauss elimination method does not fail even if one of the pivot elements is equal to zero. The above statement is

a) True b) False

ix) The percentage error if 1/3 is approximated by 0.333 is

a) 0.1 b) 0.11

c) 0.01 d) 1.01.

x) Which one of the following is an iterative method ?

a) Gauss-elimination

b) Gauss-Jordon

c) Gauss-Seidel

d) none of these.

xi) Which of the following is a multistep method ?

a) Euler's method

b) Predictor-corrector method

c) Taylor's series method

d) None of these.

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xii) To solve the ordinary differential equation

$3\frac{dy}{dx} + xy^2 = \sin x, y(0) = 5$, by Runge-Kutta 2nd order method, you need to rewrite the equation as

a) $\frac{dy}{dx} = \sin x - xy^2, y(0) = 5$

b) $\frac{dy}{dx} = \frac{1}{3}\left(\sin x - xy^2\right), y(0) = 5$

c) $\frac{dy}{dx} = \frac{1}{3}\left(-\cos x - \frac{xy^3}{3}\right), y(0) = 5$

d) None of these.

GROUP – B

(Short Answer Type Questions)

Answer any *three* of the following. $3 \times 5 = 15$

2. From the following table, find the value of $f(1.5)$ by Newton forward interpolation formula :

x	1	2	3	4	5	6
$f(x)$	10	15	20	25	30	35

3. Solve the following system by Matrix Inversion Method :

$2x + y + z = 4$

$3x + 2y + 3z = 8$

$x + 4y + 9z = 14$

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4. Using Euler's method obtain the solution of $\frac{dy}{dx} = x - y$, $y(0) = 1$, and $h = 0.1$ at $x = 0.5$

5. Find out the root of the following equation using Regula Falsi method :

$xe^x - \cos x = 0$, that lies between 0 and 1 (correct to four decimal places). http://www.makaut.com

6. Evaluate the approximate value of $\int_0^1 \frac{dx}{1+x^2}$ by Simpsons $\frac{1}{3}$ rule for 4 sub-interval, correct up to 4 decimal places.

GROUP – C

(Long Answer Type Questions)

Answer any *three* of the following. $3 \times 15 = 45$

7. a) Using appropriate formula find f (0.29) from the following table : 7

x:	0.20	0.22	0.24	0.26	0.28	0.30
$f(x)$:	1.6596	1.6698	1.6804	1.6912	1.7024	1.7139

b) Compute the value of $\int_{1.2}^{1.6} (x + \frac{1}{x})dx$, taking $h = 0.05$ correct up to five decimal places by using (i) Trapezoidal rule, (ii) Simpson's $\frac{1}{3}rd$ rule. 4 + 4

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8. a) Solve by Gauss-Seidel iterative method :

$$3x + 9y - 2z = 11$$

$$4x + 2y + 13z = 24$$

$$4x - 2y + z = -8$$

Correct up to four significant figure. 8

b) Find a $+ve$ root of $x + \ln x - 2 = 0$ by Newton. Raphson method, correct up to six significant figure. 7

9. a) Compute y (0.8), by Runge-Kutta method correct up to five decimal places from the equation $\frac{dy}{dx} = xy, y(0) = 2$ taking $h = 0.2$. 7

b) Solve by LU decomposition method :

$$2x - 6y + 8z = 24$$

$$5x + 4y - 3z = 2$$

$$3x + y + 2z = 16$$ 8

10. a) Find $f(8)$ using Newton's divided difference formula given that

x:	4	5	7	10	11	13
$f(x)$:	48	100	294	900	1210	2028

7

b) Derive the expression of Newton's forward interpolation formula where the function $f(x)$ is known for $n + 1$ distinct equispaced arguments. 8

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11. a) Solve be matrix inversion method :

$$2x - 3y + 10z = 3$$

$$-x + 4y + 2z = 20$$

$$5x + 2y + z = -12$$ 8

b) Use modified Euler method to find the value of $y(0.02)$ by taking $h = 0.01$ of the differential equation $\frac{dy}{dx} = x^2 + y$, given that $y(0) = 1$. 7

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